

Reproducing “*No cosmological constraints on dark photon dark matter from resonant conversion*”

(arXiv:2510.13956) with Entity

Implementation on Lonestar6 and comparison with the published runs

September 15, 2026

Abstract

This note documents a reproduction of the particle-in-cell (PIC) results of Hook, Huang & Shalaby (arXiv:2510.13956v1) using the ENTITY PIC framework in place of the authors’ SHARP code. Part I covers the implementation: a dedicated problem generator, a deterministic (“quiet”) particle loader written for this work, the four build configurations required on TACC Lonestar6, and a measured performance characterisation across CPU, single-GPU and multi-GPU execution. Part II documents every production run, the exact settings used, and a point-by-point comparison with the parameters and results stated in the paper. Both resonance cases of the paper’s Fig. 2 are reproduced; the Landau–Zener family is in progress. All performance and physics numbers quoted are measured on Lonestar6 unless explicitly labelled as an estimate.

Contents

1	Scope and conventions	2
I	Implementation	3
2	Problem generator	3
3	Deterministic (“quiet”) particle loading	3
3.1	What a quiet start must actually kill	3
3.2	Construction	4
3.3	Validation	4
3.4	What the quiet start does <i>not</i> buy	5
4	Building on Lonestar6	5
4.1	Environment and prerequisites	5
4.2	Configure	5
4.3	Compile	6
5	Running on Lonestar6	7
5.1	Queues	7
5.2	Measured performance	7
5.3	Per-step cost doubles after saturation	8
5.4	Checkpointing	8
5.5	Queue behaviour dominates scaling	8

5.6 Operational traps	8
II Runs, settings, and comparison with the paper	9
6 Parameters taken from the paper source	9
7 Preparatory and calibration runs	9
8 The two resonance cases (paper Fig. 2)	10
9 Landau–Zener runs	10
10 Corrections to earlier analysis	11
11 Status and remaining work	11

1 Scope and conventions

The paper studies the resonant conversion of dark-photon dark matter (DPDM) into a plasma excitation in the early-universe electron–ion plasma, and shows that *nonlinear* plasma dynamics saturate the conversion far below the linear-theory expectation, weakening the resulting cosmological limits. The evidence is a set of 1D–1V electrostatic PIC simulations.

Stripped of the cosmology, the numerical problem is small and sharply defined:

- 1D electrostatic, periodic box, $N_x = 1000$, $L = 40 c/\omega_p$, $\Delta x = 0.04 c/\omega_p \simeq \lambda_D$;
- electrons and ions with the *real* mass ratio $m_i/m_e = 1836$, both evolved;
- $k_B T_e = k_B T_i = 10^{-3} m_e c^2$;
- the dark photon enters as a *spatially uniform* ($k = 0$), time-oscillating external electric field $E_{\text{ext}}(t) = A_0 \cos(\omega t)$ acting on the particle Lorentz force *only* — it is never a source term in Maxwell’s equations. All feedback is through the deposited current.

Code units. Throughout, $\omega_{pe} = 1$, $c = 1$, $n_e = n_0$, $m_e = 1$. Energies are in $n_e m_e c^2$, times in ω_p^{-1} , wavenumbers in ω_p/c . The drive amplitude in code units equals the paper’s: $A_0 = v_q^D/c = (v_q^D/v_{\text{th}}^e)\sqrt{10^{-3}}$.

A note on what is measured. This document distinguishes carefully between measured and estimated quantities, because several early estimates in this project turned out to be wrong by large factors and were corrected only after production runs exposed them. Section 10 lists those corrections explicitly.

Part I

Implementation

2 Problem generator

All problem-specific code lives in `pgens/dpdm/pgen.hpp`. It provides three things.

External field. An `ExternalFields` hook returning a uniform e_{x1} , applied to *both* species; the pusher scales by q/m per species, so the ions automatically receive $1/1836$ of the electron kick. Because the drive is uniform in space, the cosine is evaluated once per species per step on the host and only the scalar is handed to the kernel — not once per particle.

Two drive families. Selected by `setup.drive`:

$$\text{resonance: } \phi(t) = \omega t, \quad \omega = \omega_p \text{ fixed}, \quad (1)$$

$$\text{Landau-Zener: } \phi(t) = \omega_0 t + \frac{1}{2} \dot{\omega} t^2. \quad (2)$$

The swept phase is integrated *analytically*. Sampling $\omega(t)$ per step and multiplying by t accumulates a spurious chirp: the naive form deviates by up to $0.86 A_0$ over the sweep. The analytic form was verified against the closed expression to 2.8×10^{-14} (double precision, i.e. exact).

A custom diagnostic. `CustomStat` exposes `e_ext`, the instantaneous drive, so amplitude and phase can be checked against the plasma response in the same CSV. Note that `CustomStat` receives time one timestep behind the `time` column in the same row (`engine.hpp` passes `time - dt` to the writer); this is a diagnostic bookkeeping offset only — the pusher’s centring is correct, as the drive-normalisation measurement in §7 confirms.

3 Deterministic (“quiet”) particle loading

The paper initialises “all particles equally spaced, with ions and electrons colocated throughout the computational domain,” which ensures $\epsilon_{\text{noise}} = 0$ at $t = 0$. ENTITY’s stock injector places particles at *random* sub-cell positions and has no quiet-start option, so one was written for this work.

3.1 What a quiet start must actually kill

There are three independent noise channels, and the phrase “equally spaced” only obviously addresses one:

1. **Charge density.** $\rho(x, 0) \neq 0$ seeds $E(x, 0) \neq 0$. Killed by co-locating each electron with an ion. ENTITY’s *random* start already does this, so this channel was never the problem.
2. **Total density.** $n_{\text{tot}} = n_e + n_i$ fluctuates at $\sqrt{2/N_{\text{ppc}}}$ even with perfect co-location, because both species fluctuate together. Since $\omega_p \propto \sqrt{n}$, this is a spread in the local resonant frequency — the channel that physically matters.
3. **Velocity space.** Even on a perfect lattice, randomly sampled velocities produce current fluctuations that regenerate E within a plasma period and re-seed channel 2 dynamically.

A loader that fixes 1 and 2 but not 3 looks perfect at $t = 0$ and then relaxes to the random-start floor within a few ω_p^{-1} .

3.2 Construction

The essential requirement — and the one that took two attempts to get right — is that **each velocity class must itself be a uniform lattice over the box**. Then under free streaming a class merely translates, and a Δx -spaced lattice deposits *exactly* uniform density at any translation, because the shape function is a partition of unity at that spacing. Indexing by velocity class rather than by sub-cell slot:

```
c = p / ncell          // velocity class: exactly ncell particles each
i = p % ncell          // cell
phi = ((c*P0) mod ppc + 1/2)/ppc  // sub-cell offset: a function of c ALONE
x = xi_min + i + phi
u_e = sig1 * probit((c + 1/2)/ppc)
u_i = sig2 * probit((c*P2) mod ppc + 1/2)/ppc)
```

Four properties, each doing a specific job:

- `phi` depends only on `c`, so every velocity class is a perfect Δx lattice. *This is what makes the order survive streaming.*
- `P0`, `P2` coprime to `ppc` make the index maps bijections, so each cell carries the complete stratified quantile set exactly once (zero net drift, exact temperature per cell) while consecutive slots receive *non*-consecutive quantiles, so no cell holds a systematic $v(x)$ ramp.
- Ions reuse the electron positions, so the charge density vanishes bit-for-bit at $t = 0$; their velocities use a second stride so the species are not phase-space correlated.
- Nothing references the cell index or the domain offset, so the loading is independent of how MPI decomposes the domain.

The failed first design, and why. The initial implementation tied the velocity to the sub-cell *slot* with a per-cell rotation. A given velocity then sits at a different offset in each cell, the classes shear apart as soon as particles move, and the measured result was $\delta n/n = 6.3 \times 10^{-3}$ after a *single* step, with a noise floor only $2\times$ below the random start. Getting $t = 0$ clean is easy; making the order survive is the actual problem.

Velocity normalisation. The stratified set $\{\text{probit}((j + \frac{1}{2})/N_{\text{pc}})\}$ has mean zero by symmetry but variance *below* unity, because midpoint quantiles under-sample the Gaussian tails: -1.97% at $N_{\text{pc}} = 64$, -0.64% at $N_{\text{pc}} = 200$, -0.07% at $N_{\text{pc}} = 2000$. A device-side reduction measures the realised variance and divides it out, so the loaded temperature is exact. Without this the plasma runs systematically cold at production particle counts — an invisible bias of exactly the kind a quiet start exists to remove. `probit` is Acklam’s rational approximation plus one Halley step against `erfc`, accurate to $\sim 10^{-16}$ over $q \in [10^{-8}, 1 - 10^{-8}]$.

3.3 Validation

Gate	Result	Reference
Lattice uniformity	$\delta n/n = 0.000 \times 10^0$ (exactly)	random: $1/\sqrt{N_{\text{pc}}} = 0.125$
Charge neutrality	$\max \rho_q = 0.000 \times 10^0$	—
Undriven noise floor	1.31×10^{-17}	random: 3.03×10^{-6}
No coherent seeded mode	peak/median = 10.6, power $\sim 10^{-38}$	—
Drive normalisation	$C = 0.12531$	theory $1/8 = 0.125$
8-domain MPI vs 1 domain	agreement to 1.5×10^{-12}	reduction ordering

Table 1: Quiet-start validation gates, $N_{\text{pc}} = 64$. Scripts: `validate_quiet.sh`, `check_quiet.py`, `validate_mpi.sh`.

Two traps worth recording. ENTITY’s `Rho` field is *mass* density ($1 + 1836 = 1837$); the charge density is the separate `Charge` field. And $E_x(t=0) = 0$ proves nothing, because ENTITY initialises

E to zero identically and *both* loadings co-locate the pair — the discriminator is the *total* density $n_e + n_i$.

3.4 What the quiet start does *not* buy

An undriven run at $N_{\text{pc}} = 2700$ over the full Landau–Zener duration ($\omega_p t = 2 \times 10^4, 10^6$ steps) shows the ordered state *relaxing*:

ϵ_E at $t = 0$	6.7×10^{-23}
ϵ_E plateau (reached by $\omega_p t \approx 5000$)	4.8×10^{-8}
random-start floor at the same N_{pc}	7.6×10^{-8}

The lattice starts at round-off and relaxes to $\sim 62\%$ of the thermal shot-noise floor. This is the paper’s own statement (“it takes approximately $t\omega_p \sim 3\text{--}4$ for the driven energy to be converted into de-coherent high- k noise”) seen from the undriven side. **Consequence: a quiet start does not permit a reduction in N_{pc} .** It buys a clean $t = 0$ and nothing more; particle counts must still be sized from the paper’s shot-noise criterion.

The same run bounds numerical heating over 10^6 steps with ordered loading:

$$\Delta(k_B T_e)/k_B T_e = -0.0162\%, \quad \Delta(k_B T_i)/k_B T_i = +0.0084\%.$$

An ordered lattice does not drift. This was the gate for the Landau–Zener runs, where a slow T_e creep would shift ω_p and masquerade as physics.

4 Building on Lonestar6

This section is the full recipe: the four build directories used in this work can be reconstructed from scratch by following it in order. The directories themselves are not version-controlled — they total ~ 1.2 GB of object files, fetched dependency sources and architecture-specific binaries, and are valid only on this machine.

4.1 Environment and prerequisites

All four builds are configured and compiled **on a login node**. ENTITY fetches Kokkos and ADIOS2 with CMake’s `FetchContent` at configure time, and LS6 compute nodes have no outbound network: the same commands on a compute node fail in the clone step.

```
module load gcc/13.2.0    # switches the hierarchy; brings impi/21.12 with it
module load cuda/12.8    # CUDA builds only -- LS6 defaults to cuda/11.4
```

`gcc/13.2.0` is not optional. The binaries link against GCC 13’s `libstdc++`; the default LS6 environment provides `gcc/9.4.0`, under which the executable dies at load time with `GLIBCXX.3.4.31 not found before` it reads its input file. The module must therefore also be loaded in every job script, not only at build time. Loading it switches the module hierarchy to `/opt/apps/gcc13.2/modulefiles` and pulls in `impi/21.12` (Intel oneAPI MPI 2021.12, `mpigxx`), which is the MPI the two parallel builds pick up. CUDA must be requested explicitly: the site default `cuda/11.4` is too old for the fetched Kokkos, and the builds here used `cuda/12.8` (`nvcc` at `/opt/apps/cuda/12.8/bin/nvcc`).

4.2 Configure

Four configurations are used, differing only in backend. All are configured from the repository root:

```

IGN=-DCMAKE_IGNORE_PATH=$PWD/cmake
COMMON="-D pgen=dpdm -D deposit=esirkepov -D shape_order=5 -D precision=double"
KOK="-D Kokkos_ENABLE_OPENMP=ON -D Kokkos_ENABLE_SERIAL=OFF"

# 1. CPU, OpenMP          -- validation, fast resonance case
cmake -B build-dpdm        $COMMON $KOK -D Kokkos_ARCH_ZEN3=ON $IGN
# 2. CPU + MPI            -- multi-node fallback
cmake -B build-dpdm-mpi    $COMMON $KOK -D Kokkos_ARCH_ZEN3=ON -D mpi=ON $IGN
# 3. CUDA, single GPU     -- the production path
cmake -B build-dpdm-cuda   $COMMON -D Kokkos_ENABLE_CUDA=ON \
-D Kokkos_ARCH_AMPERE80=ON $IGN
# 4. CUDA + MPI, multi-GPU -- LZ 0.1, long-term stability
cmake -B build-dpdm-cuda-mpi $COMMON $KOK -D mpi=ON -D gpu_aware_mpi=OFF \
-D Kokkos_ENABLE_CUDA=ON -D Kokkos_ARCH_AMPERE80=ON $IGN

```

Kokkos_ARCH.ZEN3 targets the AMD EPYC 7763 CPU nodes and Kokkos_ARCH.AMPERE80 the A100s; configuration 3 omits \$KOK because a CUDA backend does not need a threaded host backend for this workload. Each `cmake -B` re-fetches and re-builds its own private copy of Kokkos and ADIOS2 under `<build>/_deps`, which is why four directories cost ~ 1.2 GB rather than four small binaries.

Historical note. Configuration 1 was in fact first configured with `-D adios2_DIR=$PWD/build/_deps/adios2-build` reusing the ADIOS2 that an earlier unrelated `build/` directory had already fetched. That works, but it is not reproducible from a clean checkout — there is no `build/` to point at — so the form given above, using \$IGN like the other three, is the one to follow. The two are equivalent: both end up at ADIOS2 2.11.0.

4.3 Compile

```

cmake --build build-dpdm      -j 8
cmake --build build-dpdm-mpi  -j 8
cmake --build build-dpdm-cuda -j 8
cmake --build build-dpdm-cuda-mpi -j 8

```

Each produces its executable at `<build>/src/entity.xc`; that path is what the job scripts in `pgens/dpdm/` invoke. The CUDA configurations take substantially longer than the CPU ones — every kernel is compiled twice, to PTX and to Ampere SASS. Keep `-j` modest: LS6 login nodes are shared, and a large `-j` on an `nvcc` build is the usual way to get a job killed by the login-node watchdog.

Verify before trusting a build. Configure-time options fail silently here (Trap 2 below), so check the cache rather than assuming:

```

grep -E " ^Kokkos_ENABLE_(OPENMP|SERIAL|CUDA):BOOL" build-*/CMakeCache.txt

```

Trap 1: the ADIOS2 shim shadows its own package. `find_package(adios2 QUIET)` matches the repository’s own `cmake/adios2Config.cmake`, which only sets cache variables and exports neither a version nor any targets. CMake nevertheless sets `adios2_FOUND` and the fetch branch never runs. The symptom is ADIOS2: `v` with a path pointing into `entity/cmake`, followed by Target `"entity.xc"` links to `adios2::cxx_mpi` but the target was not found. Excluding that directory via `CMAKE_IGNORE_PATH` restores the intended fetch. This is an upstream bug: the shim shadows the package it exists to configure.

Trap 2: CMAKE_IGNORE_PATH also disables Kokkos OpenMP. The same exclusion blocks the repository’s Kokkos settings file, and a freshly fetched Kokkos defaults to **Serial only**. A Serial build is externally indistinguishable from a threaded one and runs $\sim 112\times$ slower (NLWP = 2, 99.5% CPU). This cost a 40-minute walltime overrun and two incorrect diagnoses (SLURM -c, then ibrun binding — both innocent). This is why the cache check at the end of the previous subsection is mandatory and not advisory.

Trap 3: ADIOS2 target rename. ADIOS2 2.11 renamed the C++ bindings target from `cxx11` to `cxx` (and `cxx11_mpi` to `cxx_mpi`). ENTITY hardcodes the 2.11 names, so TACC’s `adios2/2.10.2` module cannot link even though it does provide `ADIOS2_HAVE_MPI`. Fetching 2.11.0 (ENTITY’s pinned tag) is the clean route, and requires a login node — compute nodes have no outbound network.

GPU-aware MPI. Disabled (`gpu_aware_mpi=OFF`). It would require `mvapich2-gdr-cuda12`, a different MPI stack from the `impi` in the `gcc13` hierarchy. The halo here is a few ghost cells of three field components, so host staging is free at this problem size.

5 Running on Lonestar6

5.1 Queues

Partition	MinNode	MaxNode	MaxWall	MaxNodePU	MaxJobsPU
development	1	8	02:00:00	8	1
normal	1	64	2-00:00:00	64	20
gpu-a100	1	8	2-00:00:00	12	8
gpu-a100-dev	1	2	02:00:00	2	1
gpu-a100-small	1	1	2-00:00:00	3	3

Anything under 2 h and ≤ 8 nodes belongs in `development`: turnaround there was measured at 9 s and 70 s for jobs that never started at all in `normal`. `MaxJobsPU = 1` only means short jobs run back-to-back, which is still far faster than waiting on `normal` priority. `gpu-a100` nodes carry three A100-40GB each and bill at weight 128; `gpu-a100-small` nodes carry one and bill at 32, which makes them the correct target for single-GPU production.

5.2 Measured performance

Configuration	ms/step at $N_{pc} = 2 \times 10^5$	Note
1 CPU node (128 cores)	~ 950	steady state; rises +48% from 640
8 CPU nodes, MPI	135	4.94 $\times$ speedup, 69% efficiency
1 $\times$ A100	265 $\rightarrow$ 516	3.6 $\times$ a full CPU node
3 $\times$ A100 (one node)	128	2.07 $\times$ over 1 GPU, 69% efficiency

Table 2: Measured on 4×10^8 particles, $N_x = 1000$.

A single A100 outperforms an entire 128-core node by 3.6 $\times$. The feared collapse from deposition atomics on a 1000-cell grid did not occur: going from $N_{pc} = 2 \times 10^4$ to 2×10^5 — a tenfold increase in particles per cell, and therefore in atomic contention per cell — cost only a factor 1.42 (0.444 $\rightarrow$ 0.630 ns per particle-step).

5.3 Per-step cost doubles after saturation

This is the most expensive lesson of the project and it governs all walltime planning.

On a single A100 at $N_{\text{pc}} = 2 \times 10^5$ the cost is 265 ms/step while the plasma is laminar and 516 ms/step once it saturates and becomes turbulent — particles disorder, and the memory locality the early rate depended on is lost. A 2000-step benchmark ($\omega_p t = 40$) showed a dead-flat 265 ms and was taken as evidence of stability. It was nothing of the kind: the run saturates at $\omega_p t \approx 2000$, so the benchmark never left the laminar phase. The slow resonance case consequently overran a 48 h wall at 88% completion and required a checkpoint restart.

Rule: benchmark across the physics transition, or budget a factor two.

5.4 Checkpointing

Checkpoints are written by `[checkpoint] interval_time` with `keep` retained, and a run resumes with the `-resume` flag (aliases `-continue`, `-restart`, `-checkpoint`); ENTITY selects the checkpoint automatically. This recovered the slow resonance case after its walltime overrun.

The domain count is locked by the checkpoint format: an N -domain checkpoint can only be resumed by N domains. A single-GPU checkpoint cannot be resumed by a 3-GPU MPI binary. The GPU/rank count must therefore be decided before the first submission of any run that may need to restart.

5.5 Queue behaviour dominates scaling

On a saturated machine, queue wait swamps compute. Multi-GPU benchmark jobs requesting 2 and 4 nodes sat unscheduled for 30 hours and were eventually cancelled, while single-node jobs ran within minutes. A 12-GPU configuration with a theoretical 6 h runtime is worthless if it waits 30 h for four contiguous nodes. **Prefer fewer nodes than the scaling curve alone would suggest.**

5.6 Operational traps

1. `ppc0` must be a TOML *float*. `ppc0 = 2000` parses as an integer and ENTITY aborts with `toml::value::as_floating(): bad_cast`. Write `2000.0`. This caused two separate job failures.
2. Default logging is two firehoses. `<name>.log` is a verbose kernel trace at ~ 4 kB *per step* (governed by `log_level`), and `<name>.out` is a progress bar redrawn every step (governed by `interval`). Defaults produced 107 MB of text beside 75 kB of science on a 2×10^4 -step run, written synchronously to the parallel filesystem. Set `log_level = "WARNING"`, `interval = 200`, `colored_stdout = false`.
3. `[output.spectra]` defaults to *on*. Disable it explicitly.
4. ENTITY races on output-directory creation under MPI: every rank calls `mkdir` on `<simulation.name>/` and the losers abort with `filesystem error: cannot create directory: File exists, exit 255`, within seconds — which resembles a suspiciously fast run. Pre-create `<rundir>/<name>/` in the batch script.
5. The stats CSV header uses literal names (`E1^2`, `J.E`); the enum lookup is lowercase `"j.e"` and `"JdotE"` aborts. T00 alone gives the summed value — per-species requires `T00_1`, `T00_2`.
6. Per-species stats force a moment reduction over all particles; at `interval_time = 0.1` this doubled wall-clock. Keep it ≥ 1.0 .
7. `blocking_timers = true` is the first move on any performance surprise. It prints a per-kernel breakdown with ns-per-particle, and answered in nine seconds a question that two rounds of speculation had got wrong.

Part II

Runs, settings, and comparison with the paper

6 Parameters taken from the paper source

All values below are quoted from `main.tex` in the arXiv source package, not inferred from the text or figures.

Quantity	Paper value	This work
Grid	$N_x = 1000, L = 40 c/\omega_p$	identical
Cell size	$\Delta x = 0.04 c/\omega_p$	identical
Debye length	$\lambda_D \approx 0.033 c/\omega_p$	identical
Temperatures	$k_B T_e = k_B T_i = 10^{-3} m_e c^2$	identical
Mass ratio	$m_i/m_e = 1836$	identical
Thermal speed	$v_{th,e}^2 \equiv \langle (v_x - \langle v_x \rangle)^2 \rangle = k_B T/m$	identical (1V)
Thermal energy density	$5 \times 10^{-4} n_e m_e c^2$	4.9963×10^{-4} measured
Deposition	5th-order splines	<code>esirkepov, shape_order=5</code>
Pusher	Vay (2008)	<code>Vay</code>
Timestep	$\Delta t \approx 0.02/\omega_p$	$CFL = 0.5 \Rightarrow 0.02$
Initial loading	equally spaced, co-located	<code>§3</code>
Shot noise	$\epsilon_{noise} = L^2/(12N_p N_x)$	agrees to $3.1 \times$
Resonance N_{pc}	2×10^2 and 2×10^5	identical
LZ sweep	$\omega(t) = 0.8\omega_p + 0.1\omega_p t/5000$	identical
LZ amplitudes	$v_q^D/v_{th}^e = 1.0$ and 0.1	identical
LZ N_{pc} criterion	$\bar{t}_{noise} \leq 0.02$	$2.7 \times 10^3, 2.7 \times 10^5$

Table 3: Paper parameters and the corresponding settings used here.

Two points deserve emphasis.

The 1V definition. The paper defines the thermal speed from the *one-dimensional* x -component variance, $v_{th,e}^2 = \langle (v_x - \langle v_x \rangle)^2 \rangle = k_B T/m$. ENTITY’s Maxwellian is isotropic 3V. Because there are no transverse forces in a 1D electrostatic run, the perpendicular degrees of freedom merely carry constant energy; including them would dilute T_e by a factor three. The quiet loader therefore sets $u_{x2} = u_{x3} = 0$ (`one_v = true`), matching SHARP exactly and removing the 3V-versus-1V correction from every downstream comparison. The paper’s quoted thermal energy density 5×10^{-4} matches the 1V value and is used as the reference line throughout.

The Landau–Zener crossing time. With $\omega_0 = 0.8$ and $\dot{\omega} = 2 \times 10^{-5}$, resonance ($\omega = \omega_p$) is crossed at $\omega_p t = 10^4$. The paper’s statement that “nonlinearity kills the resonant transfer around $\omega_p t = 5000$ ” refers to $\omega = 0.9\omega_p$, i.e. *before* the crossing — off-resonant excitation, not the crossing itself.

7 Preparatory and calibration runs

Before the production runs, a sequence of measurements pinned the unit conventions and validated the drive. These are summarised here because several production choices depend on them.

- **Units.** `skindepth0 = 1` with `densities = [2.0]` gives $n_e = n_0$ exactly, hence $\omega_{pe} = 1$. Confirmed by `T00.1` rest-mass part = 1.0000.
- **Energy conservation.** Over 2×10^4 steps, $\Delta T^{00} = -\frac{1}{2}\Delta\langle E^2 \rangle$ with $r = -0.996$; no grid heating.

- **ω_p calibration.** A five-point detuning scan ($\omega = 0.96\text{--}1.04$) is symmetric about $\omega = 1$ and follows the linear-theory sinc^2 envelope; the on-resonance case grows as t^2 across two decades in energy while every detuned case remains bounded.
- **Drive normalisation.** Least squares over $10 < \omega_p t < 40$ gives $C = 0.12432$ against Eq. A25’s $1/8 = 0.125$ (0.5%). The ion response ratio is 5.70×10^{-4} against $1/1836 = 5.45 \times 10^{-4}$, confirming the drive reaches both species with the correct q/m .

A factor-of-two trap. It is the *sum* $\epsilon_E + \Delta K E_e$ that obeys the t^2 law, not the field energy alone. The resonantly driven Langmuir wave equipartitions, so the field carries only half the wave energy on cycle average and swings between 3% and 85% of it instantaneously at $2\omega_p$. The paper’s linear-theory line must be compared against the sum.

8 The two resonance cases (paper Fig. 2)

Both run to $\omega_p t = 10^4$ with quiet loading, 1V, on a single A100.

Case	N_{pc}	A_0	C	peak $\epsilon_E/\epsilon_{\text{th}}$	final $\Delta K E_e/\epsilon_{\text{th}}$
fast, $v_q^D/v_{\text{th}}^e = 3 \times 10^{-2}$	2×10^2	9.48683×10^{-4}	0.1264	43.6 (at $\omega_p t = 553$)	69.6
slow, $v_q^D/v_{\text{th}}^e = 1 \times 10^{-3}$	2×10^5	3.16228×10^{-5}	0.1237	2.1 (at $\omega_p t = 3694$)	2.4

Table 4: Production resonance runs. C is the measured coefficient of $\epsilon_E + \Delta K E_e = C A_0^2 (\omega_p t)^2$; linear theory gives $1/8 = 0.125$. $\epsilon_{\text{th}} = 5 \times 10^{-4}$ is the 1V electron thermal energy density.

Initial temperature. $k_B T_e(0) = 4.9963 \times 10^{-4}$ against the 1V target 5.0×10^{-4} . The -0.075% residual is *not* loading error but the relativistic definition $T^{00} = \sum w m \gamma$, whose next term is $-\frac{3}{8}\langle u^4 \rangle = -0.075\%$. The ions, 1836 times heavier and far more non-relativistic, land at 4.999998×10^{-4} , which confirms the interpretation.

The regime distinction. The paper separates slow and fast growth at $v_q^D/v_{\text{th}}^e \sim (m_e/m_p)^{1/2}/2 \approx 0.0117$. Both of our cases overshoot the thermal energy in the *field* energy (43.6 versus 2.1), but the lasting electron heating differs by a factor ~ 29 : $O(10^2)$ for the fast case, versus a slow case whose electrons are barely changed — which is the paper’s own phrasing for the Landau–Zener result (“the kinetic energy of the electrons is not even changed by a factor of 2”). **The discriminator is the final electron heating, not the peak field energy.**

Saturation is delayed relative to random-start runs. With the noise floor eleven orders of magnitude lower at $t = 0$, daughter waves grow from a far smaller seed, so saturation occurs later. The fast case peaks at $\omega_p t = 553$ rather than the ~ 110 ($t_{\text{sat}} + 1/\omega_{pi}$) seen in earlier random-start scans. This is expected behaviour of a correct quiet start, and it means random-start timings are not a valid comparison.

Cost. The slow case required ~ 51 GPU-hours in total: 48 h to reach 88% completion, then a 14.5 h checkpoint restart — a direct consequence of the post-saturation slowdown of §5.3.

9 Landau–Zener runs

The Landau–Zener family is the physically realistic case: in cosmology it is $\omega_p(t)$ that drifts through the dark-photon mass, and the duration of the real transition is $\sim 10^{10}$ plasma periods, far beyond any simulation. The paper instead sweeps $m_{A'}(t)$ at a numerically tractable rate corresponding to $\omega_p/H \approx 7 \times 10^4$.

Run	v_q^D/v_{th}^e	N_{pc}	Resources	Status
Heating gate (undriven)	—	2.7×10^3	1 A100, 2 h	complete: T_e drift -0.016%
LZ high amplitude	1.0	2.7×10^3	1 A100, ~ 2 h	running
LZ low amplitude	0.1	2.7×10^5	3 A100, ~ 81 h chained	written, gated

Particle counts. The Landau–Zener noise criterion differs fundamentally from the resonant one. Before resonance the drive is a *bounded* forced oscillation of amplitude $A_0/(1 - \omega^2/\omega_p^2) = 2.78A_0$ at $\omega = 0.8\omega_p$, not a growing one, so there is no time at which the signal outgrows the noise. The paper requires $\bar{t}_{\text{noise}} = L/(A_0\sqrt{3N_p N_x/2}) \leq 0.02$ — driven energy fifty times the shot noise — which gives $N_{\text{pc}} = 2.7 \times 10^3$ and 2.7×10^5 for the two amplitudes. Note that the quiet start does *not* relax this requirement (§3.4).

Why the low-amplitude case needs three GPUs. At $N_{\text{pc}} = 2.7 \times 10^5$ the run holds 5.3×10^8 particles (≈ 31 GB), which is tight on a single 40 GB A100 and comfortable at 10.5 GB per GPU across three. Combined with the post-saturation slowdown, a single GPU would need ~ 168 h; three reduce this to ~ 81 h. Because the checkpoint domain count is locked, this must be chosen at first submission.

10 Corrections to earlier analysis

Three conclusions reached early in this project were wrong and propagated into planning before being caught. They are recorded because each was confidently held.

1. **“The shot-noise floor is $650\times$ below Eq. B4.”** False — a units error. N_p in $\epsilon_{\text{noise}} = L^2/(12N_p N_x)$ is the *total* macroparticle count, which the paper states explicitly; it was read as per-cell. Correctly evaluated, Eq. B4 agrees with measurement at a flat $3.1\times$ across three decades of N_{pc} . The formula is sound and should be used to size particle counts.
2. **“The quiet start makes the noise requirement irrelevant.”** False. The paper uses a quiet start *and* still requires large N_{pc} , because the ordered state relaxes to the thermal shot-noise floor within $\omega_p t \approx 5000$ (§3.4).
3. **“Per-step cost is stable.”** False, and expensive. It roughly doubles once the plasma saturates. The 2000-step benchmark that appeared to establish stability covered only the laminar phase — 0.4% of the production run, and entirely before the physics transition.

A common thread: each was an extrapolation from a measurement taken outside the regime where it was then applied. The cost model itself was calibrated at $N_{\text{pc}} \leq 8000$ and subsequently failed in both directions — predicting 8 minutes for a run that took 41, and 97 seconds of stepping for one that exceeded 40 minutes.

11 Status and remaining work

Complete. Both resonance cases of Fig. 2, the quiet-start implementation and its validation (including MPI decomposition independence), the numerical-heating gate at Landau–Zener duration, and the full performance characterisation.

In progress. Landau–Zener at $v_q^D/v_{\text{th}}^e = 1.0$; the 0.1 case is written and gated on it.

Open.

- **No comparison against the published figures.** Every number here is internally consistent and matches the paper’s *text*, but the figure PDFs in the source package have not been digitised. The peak $\epsilon_E/\epsilon_{\text{th}}$ and its timing are the sharpest available discriminators.
- **Long-term stability** ($\omega_p t \rightarrow 8 \times 10^4$), the paper’s strongest claim, is untested. At $N_{\text{pc}} = 2 \times 10^5$ it is roughly eight times the slow resonance case and would require multi-GPU execution with chaining.
- **2D cross-check** not run; 1+1D sufficiency is assumed on the paper’s word.
- **Collisions.** ENTITY, like SHARP, is collisionless. The paper treats the final ion-acoustic damping analytically via τ_{ei} . This matches the paper’s regime, but “persists indefinitely” should be read as “persists on collisionless timescales”.